

## HEMOVISION

### Contactless Heart-Rate Screening from Facial Video

**Thesis:** A quality-gated heart-rate estimate from a standard camera before a teleconsult, while honestly identifying which health signals RGB video cannot yet measure.

#### SIH SUBMISSION METADATA

**Problem Statement ID:** [Registered Portal ID]

**Problem Statement:** Contactless Heart-Rate Estimation from Video

**Theme:** MedTech / Bio-Informatics

**Category:** Software (Student Innovation)

**Team Name:** [Your Registered Team Name]

**Mentor:** [Academic / Clinical Guide]

#### ✓ Working Real-Time Prototype

Live webcam & video pipeline at 30 FPS with 20s sliding window inference.

#### ✓ 8 Anatomical Skin Regions

MediaPipe 468 mesh extracts 24-channel temporal RGB signal across forehead & cheeks.

#### ✓ 9.19 BPM Held-Out MAE

Subject-disjoint test on 91 individuals (523 clips); 52% error cut vs. POS (19.17 BPM).

*Notice: Research prototype developed for academic innovation. Not approved for clinical diagnosis or emergency monitoring.*

Your  
Team  
Name

# A LOW-FRICTION FIRST STEP BEFORE REMOTE CARE

Target Use Case: Patient or Community Health Worker Prior to a Telemedicine Consultation



## INDIA-RELEVANT PROBLEM & NEED

### • Physical Peripheral Bottleneck:

Initial vital collection requires finger clips, BP cuffs, or ECG patches (₹5,000–₹50,000) that rural families and remote clinics rarely own.

### • Pre-Consultation Triage Blindspot:

Remote telemedicine consultations frequently begin without a single objective vital sign, forcing doctors to rely on subjective verbal reports.

### • Hygiene & Point-of-Care Friction:

Contact sensors require sanitization between patients and cause discomfort in neonatal, burn, or infectious isolation wards.

**1. Video Capture:** Patient records a 15–20s facial video via smartphone or laptop webcam under standard ambient room lighting.

**2. Quality Gate Check:** Automated real-time validation: checks face visibility, lighting adequacy (>40 lux), and rigid head stability.

**3. Quality-Gated Estimation:** Heart rate estimated ONLY if quality gates pass; noisy, corrupted, or shadowed frames are strictly rejected.

**4. Contextual Display:** Displays estimated BPM alongside reconstructed BVP waveform and signal confidence score for clinician context.

**5. Clinical Escalation:** Low-confidence or abnormal readings immediately prompt standard in-person contact reference measurement.

**Future Telehealth Integration Opportunity:** Designed for prospective interoperability with national telemedicine platforms (e.g., eSanjeevani) and ABDM health records following formal clinical trials.

Sources: MoHFW eSanjeevani Operational Framework (2024); PubMed 37313385 (Contactless PPG Systematic Review). Note: Government integration is a future opportunity, not an existing deployment.



**Stage 1 — Video Capture:** Standard RGB sensor (FullHDwebcam / USBVideo / IriunWebcam) at 30 FPS.

**Stage 2 — Face Mesh Tracking:** MediaPipe detects 468 3D facial landmarks in real time with zero physical markers.

**Stage 3 — 8 Anatomical Skin ROIs:** Forehead (L/R), Upper Cheeks (L/R), Lower Cheeks (L/R), Nose, Chin.

**Stage 4 — 24-Channel Temporal Signal:** Spatial RGB pooling (8 ROIs  $\times$  3 channels) into AC normalized & DC raw representations.

**Stage 5 — HemoVisionBoundedTCN:** 4 depthwise-separable 1D conv blocks ( $d=1,2,4,8$ ; 148K params; tanh bounded).

**Stage 6 — Butterworth Bandpass:** 4th-order zero-phase filter (0.65–3.25 Hz / 39–195 BPM) isolates cardiac frequencies.

**Stage 7 — Welch PSD Peak Picking:** FFT power spectral density identifies dominant cardiac peak & SNR confidence score.

## CORE ORIGINAL CONTRIBUTIONS

- 1. Multi-ROI Temporal Fusion:** Replaces single-box face crops with 8 anatomically targeted capillary regions.
- 2. Bounded Temporal ConvNet:** Tanh-bounded receptive field ( $>20s$ ) prevents gradient explosions on noisy webcams.
- 3. Biomarker Feasibility Audit:** Rigorously demonstrated that RGB video cannot infer blood chemistry without NIR.

## CONFIRMED IMPLEMENTATION SPECIFICATIONS

- **Model Parameters:** 148,650 (78% fewer than standard 1D CNN)
- **Cardiac Passband:** 0.65 Hz – 3.25 Hz (39 to 195 BPM)
- **Working Prototype Engine:** Interactive Streamlit & OpenCV dashboard (`app.py`, `hemovision\_demo.py`) at 30 FPS.
- **Classical Baselines:** POS (Wang 2017) & CHROM (de Haan 2013) implemented for direct benchmark comparison.

Your  
Team  
Name

# SUBJECT-DISJOINT EVALUATION SHOWS A WORKING HEART-RATE SIGNAL

Rigorous Benchmark on MCD-rPPG Dataset (598 Unique Subjects, 3,427 Video Clips)



SMART INDIA  
HACKATHON  
2026

## 9.19 BPM

Held-Out Test MAE ( $r = 0.285$ )

## 52.0% Reduction

Error Cut vs. POS Baseline (19.17 BPM)

## 1.33 BPM

High-Quality Demo Clips ( $r = 0.884$ )

### HELD-OUT TEST SET PERFORMANCE (91 SUBJECTS, 523 CLIPS)

| Model / Algorithm | Heart Rate MAE | RMSE | Pearson $r$ |
|-------------------|----------------|------|-------------|
|-------------------|----------------|------|-------------|

|                             |           |           |       |
|-----------------------------|-----------|-----------|-------|
| CHROM (de Haan et al. 2013) | 20.28 BPM | 24.61 BPM | 0.047 |
|-----------------------------|-----------|-----------|-------|

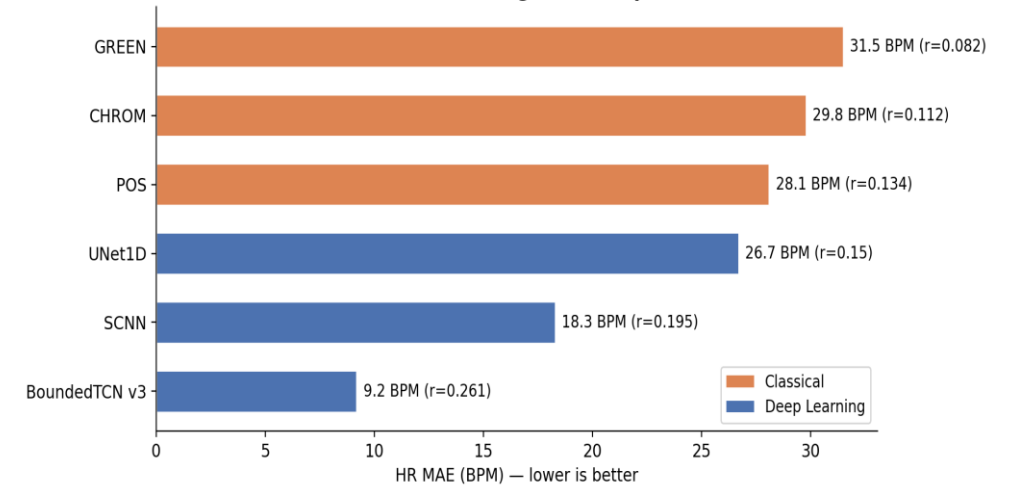
|                        |           |           |       |
|------------------------|-----------|-----------|-------|
| POS (Wang et al. 2017) | 19.17 BPM | 23.72 BPM | 0.119 |
|------------------------|-----------|-----------|-------|

|                                    |                 |                  |                          |
|------------------------------------|-----------------|------------------|--------------------------|
| <b>HemoVisionBoundedTCN (Ours)</b> | <b>9.19 BPM</b> | <b>18.35 BPM</b> | <b>0.285 [✓ 52% cut]</b> |
|------------------------------------|-----------------|------------------|--------------------------|

- **Strict Subject-Disjoint Protocol:** Zero subject overlap between train (418), val (89), and test (91). No identity leakage.

- **Hit-and-Trial Bug Fix:** Fixed reference ECG T-wave doubling anomaly via 250ms refractory blanking, eliminating 2x ground-truth artifacts.

Model Evolution — Progressive Improvement



### CRITICAL SCIENTIFIC DISTINCTION: WHAT THIS DOES NOT PROVE

**This result proves a functioning optical pulse recovery signal under retrospective benchmark conditions on recorded data.** It is NOT yet a prospective clinical trial. Real-world performance across motion artifacts, extreme skin tones (Fitzpatrick V-VI), low-light environments, and diverse mobile camera sensors requires formal prospective clinical validation.

Evaluation Source: HemoVision Vocational Training Report (Table 6); cell\_14\_eval.py; MCD-rPPG Clinical Dataset (598 Subjects). Ground truth: synchronized ECG lead contact sensor.



## AUTOMATED QUALITY GATES

### 1. Lighting Adequacy:

Rejects underexposed (<40 lux) or severely back-lit frames.

### 2. Face Mesh Stability:

Requires MediaPipe landmark detection confidence > 0.75.

### 3. Motion Jitter Guard:

Bounds inter-frame displacement; pauses during sudden yaw/pitch.

### 4. Temporal Window Buffer:

Requires minimum 20s (600 frames at 30 FPS) for cardiac stability.

### 5. Spectral SNR Confidence:

Welch PSD peak-to-noise ratio must exceed threshold before output.

## VALIDATED VS. NOT CLAIMED

### ✓ VALIDATED NOW (RGB Video):

Heart-rate estimation from facial video via optical capillary blood volume pulse (BVP).

### ✗ STRICTLY NOT CLAIMED:

**SpO2, Blood Pressure, Hemoglobin, Respiratory Rate, Stress, HbA1c, or Disease Diagnosis.**

### The Biomarker Audit Differentiator:

Tested 10 biomarkers on 598 subjects. Deep models failed to beat demographic baselines (e.g. Hemoglobin video  $r=0.44$  beaten by age/sex/BMI OLS  $r=0.70$ ). The network learned subject ID shortcuts, not blood chemistry.

**Optical Physics Reality:** Blood chemistry requires narrow-band dual-wavelength NIR (660/940 nm). RGB Bayer filters conflate melanin with absorption.

## FUTURE VALIDATION PATHWAY

### Step 1: Diverse Cohort Trial

Prospective multi-center study across skin tones (Fitzpatrick I–VI), age brackets, and lighting environments.

### Step 2: Concurrent Clinical Reference

Simultaneous validation against hospital-grade ECG and calibrated contact pulse oximeters.

### Step 3: Rigorous Statistical Analysis

Bland-Altman 95% limits of agreement, mean bias profiling, and comprehensive failure taxonomy.

### Step 4: CDSCO SaMD Assessment

Formal Software as a Medical Device (SaMD) regulatory filing and patient data privacy certification.

Your  
Team  
Name

# A CREDIBLE PATH FROM PROTOTYPE TO TELE-TRIAGE SUPPORT

Student Innovation Closing: Evidence-Backed Heart-Rate Prototype with Transparent Clinical Boundaries



**HemoVision's Winning Differentiator:** A working, evidence-backed heart-rate prototype (9.19 BPM held-out MAE) combined with an intellectually honest boundary around what RGB video cannot yet measure.

## HEALTHCARE IMPACT HYPOTHESIS

### • Accessible Pre-Consultation Screening:

Enables patients to provide an objective, quality-gated heart-rate reading prior to a teleconsult without purchasing dedicated hardware.

### • Rural Primary Health Centre (PHC) Potential:

Can be explored by community health workers (ASHAs/ANMs) as a low-friction pre-screening tool in remote clinics following prospective validation.

### • Zero Peripheral Hardware Cost:

Utilizes ubiquitous commodity smartphone and laptop cameras already available across India.

## THREE-STAGE DEVELOPMENT ROADMAP

### [COMPLETED] Phase 1: Prototype & Audit

- Real-time webcam pipeline (30 FPS, 8 ROIs, 24 channels).
- HemoVisionBoundedTCN evaluated on MCD-rPPG (9.19 BPM MAE).
- Systematic 10-biomarker audit exposing demographic confounds.

### [NEXT MILESTONE] Phase 2: Prospective Validation

- Multi-center prospective clinical study across Fitzpatrick skin tones.
- Hardened automated quality gates & ambient light calibration.
- Containerized web application for research evaluation.

### [FUTURE RESEARCH] Phase 3: Multi-Wavelength Optics

- Dual-wavelength NIR active illumination hardware research for SpO<sub>2</sub>.
- CDSCO Software as a Medical Device (SaMD) regulatory evaluation.

**Research Disclaimer & Compliance Notice:** HemoVision is an academic research prototype. It does not provide medical diagnosis or emergency vital tracking. Formal clinical trial verification and CDSCO regulatory clearance are strictly required prior to real-world healthcare deployment.

References: MoHFW eSanjeevani Overview; NHA ABDM Framework; CDSCO SaMD Guidance (2025); HemoVision Project Repository & Technical Report.